BCE315P	Fluid Mechanics Laboratory	L	T	P	C
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Pre-requisite: Fundamentals of Fluid mechanics and Hydraulic Machinery.

Course Objectives: the objectives of this course are to:

1. Understand the basic principles of fluid mechanics.
2. Explain the standard measurement techniques of fluid mechanics and their applications.
3. Compare different fluids on the basis of their properties/characteristics.
4. Illustrate the components and working principles of the Hydraulic machines- different types of Turbines and other miscellaneous hydraulics machines

Course Outcome: After successful completion of this course, the students will be able to

CO1: Utilize basic measurement techniques of fluid mechanics. and its appropriate application

CO2: Examining various types of flows, concept of fluid measurement and dimensional analysis.

CO3: Compare the results of analytical to the actual behaviour of real fluid flows, fluid properties, pressure and its measurements.

CO4: Interpret and Solve for the various hydrodynamic parameters obtained from the formulas related to the experiments.

Experiment No-1	**4 hours**
Determination of Metacentric height	
Experiment No-2	**4 hours**
Verification of Bernoulli's Theorem	
Experiment No-3	**2 hours**
Determination of C_v, C_c and C_d for an orifice	
Experiment No-4	**2 hours**
Study of the flow over Triangular notch and to find out the coefficient of discharge	
Experiment No-5	**2 hours**
Free vortex flow	
Experiment No-6	**2 hours**
Forced vortex flow	
Experiment No-7	**4 hours**
Study of hydraulic force.	
Total Practical hours	**20 hours**

Text Book(s)

1. Laboratory Manual Hydraulics and Hydraulic Machines, R V Raikar.
2. Fluid Mechanics and Hydraulic MacHines Lab Manual, Lap Lambert Academic Publishing GmbH KG.
3. Bansal R.K., "Fluid Mechanics and Hydraulic Machines".
4. R.K.Rajput, "A Text Book of Fluid Mechanics and Hydraulic Machines".

Reference Books

1. Frank. M. White, "Fluid Mechanics".
2. P.N.Modi and Seth, "Fluid Mechanics and Hydraulic Machines".
3. "Steam and Gas Turbines and Power Plant Engineering", R. Yadav, Central Publication house.
4. "Hydraulic Machines: Turbines and Pumps", G. I. Krivchenko.